

AI Acceptance in the Sharing Economy

Data Collection and Methodology — Working Report

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1 The idea

Peer-to-peer rental marketplaces — homes (Airbnb, Vrbo), cars and RVs (Turo, Getaround, RVshare, Outdoorsy), and creative gear (Fat Llama, KitSplit, Lensrentals) — run entirely on **trust between strangers**, and they are now inserting **AI and AI agents** into that relationship: automated guest screening, chatbots, dynamic pricing, automated messaging. Our question:

How do users perceive AI agents in peer-to-peer rental platforms, and what drives or erodes their trust — across rental types, and across open community discussion (Reddit) versus paying-customer reviews (Trustpilot)?

We collect two complementary corpora. **Reddit is the baseline** — where users deliberate, complain and speculate in their own words. **Trustpilot is the verification layer** — structured reviews carrying a **1–5 star rating that acts as a ground-truth sentiment label**. Where the open discussion and the labelled reviews agree we gain confidence; where they diverge we learn something about the gap between discourse and experience.

2 Data source 1 — Reddit (baseline)

Collected with `RedditExtractor` (public Reddit API). The corpus went through three passes, each fixing the previous one:

Table 1: Reddit passes. Files: `reddit_all.csv`, `reddit_targeted.csv`, `reddit_clean.csv`.

Pass	Script	Rows	Status
v1 keyword (global)	<code>scraping_data_v1.R</code>	14,090	baseline only (off-topic)
v2 subreddit-targeted	<code>01_scrape_reddit.R</code>	1,009	raw, on-topic
v2 cleaned + themed	<code>05_clean_reddit.R</code>	314	USED for analysis

Why v1 failed. A single global keyword search ("AI rental platform helpful automation trust") returned ~14,090 mostly off-topic comments (K-pop, gaming, politics); under 1% mentioned “rental”. Kept only as a transparency baseline.

How we targeted (v2). We search *inside* topic-relevant subreddits instead of all of Reddit, using **47 (subreddit, keyword) pairs** across six layers:

- **Home / stay** — `r/AirBnB`, `r/airbnb_hosts`, `r/AirBnBHosts`, `r/vrbo`, `r/ShortTermRentals`

- **Car / RV** — r/Turo, r/gigwork, r/RVing, r/GoRVing
- **Camera / film / drone** — r/photography, r/AskPhotography, r/cinematography, r/videography, r/Filmmakers, r/drones, r/dronephotography, r/DJI
- **Event tech** — r/eventproduction, r/EventProduction, r/livesound
- **AI in commerce** — r/Entrepreneur, r/SaaS, r/startups, r/automation, r/ChatGPT, r/OpenAI, r/artificial, r/singularity
- **Trust / privacy** — r/privacy, r/technology

Keywords centred on AI, automation, hosting, rental and trust (e.g. “AI”, “host AI”, “automated message”, “rental AI”, “AI agent”). Parameters: 30 threads per search, `period = "all"`, threads kept only if `comments >= 6`, capped at 400 threads; deleted/removed and sub-20-character comments dropped. The keyword that *finds* a thread is never used as its label — sentiment is derived from the text downstream.

How we balanced and cleaned. Two viral subreddits (r/Filmmakers, r/Entrepreneur) hosted threads about AI replacing humans in creative/business careers — off-topic for rental trust — so `05_clean_reddit.R` drops them and tags each remaining comment `sharing_economy` or `ai_tech`. (The scrape can also cap comments per thread at 120 to stop any single thread dominating.)

The dataset (`reddit_clean.csv`): 314 comments, 17 subreddits, 38 threads, Oct 2012 – May 2026, mean length 239 characters. Themes are close to balanced (below); the top subreddits are r/ChatGPT (81), r/photography (76), r/privacy (56), r/GoRVing (38), r/AirBnB (22), r/dji (13).

Table 2: Reddit theme balance.

Theme	Comments	Share
ai_tech	146	46%
sharing_economy	168	54%

3 Data source 2 — Trustpilot (verification)

Trustpilot gives what Reddit cannot: a **1–5 star rating on every review** — a ground-truth label we did not have to infer. We can validate a sentiment lexicon against these stars, then carry the validated lexicon to the unlabelled Reddit comments (convergent validity).

How we scraped it. Trustpilot blocks naive scraping, so we used the Apify actor `casper11515/trustpilot-reviews-scraper` via the **Apify web UI, once per company**, downloading each result CSV into `data/trustpilot_raw/`. `02_merge_trustpilot.R` then merges them (robust to BOM/UTF-8 export issues), normalises columns, and keeps English reviews with real text and unique IDs -> `data/trustpilot_reviews.csv`.

The dataset: 3,912 unique English reviews across 7 platforms, spanning Jul 2019 – May 2026. Each row carries a star rating, title, body, reviewer country and (where present) the company reply.

Table 3: Trustpilot reviews per platform, with mean star rating.

Platform	Reviews	Mean star
Turo	1482	4.62
Fat Llama	835	4.11
RVshare	773	3.98
Outdoorsy	751	3.68

Platform	Reviews	Mean star
Lensrentals	35	2.94
Getaround	31	4.35
KitSplit	5	1.00

Ratings collapse to **706 negative** (1-2 stars), **139 neutral** (3), **3067 positive** (4-5). The strong positive skew is typical of opt-in review sites — we down-sample the positive class for balanced classification.

4 Fit and next steps

The two sources triangulate: compute lexicon sentiment on Trustpilot text, validate it against the star rating, apply the validated lexicon to Reddit, then compare which *topics* drive negative sentiment in each. Agreement across two structurally different corpora is stronger evidence than either alone.

Planned analysis (owners to be assigned by the team):

- **Preprocessing + topic modelling** — clean, lemmatise, tokenise (1- and 2-grams) both corpora; fit 3–4 LDA topics per theme (small corpus -> cap at four) plus a seeded LDA on trust words.
- **Visualisation + report** — top-word bar charts, comparison/commonality word clouds, co-occurrence networks, a combined sentiment time-series; fold into the final paper.
- **Sentiment + embeddings + synthesis** — AFINN/NRC sentiment validated against Trustpilot stars, NRC emotion profiles, GloVe neighbours of *trust* / *AI* / *host*, mapped to trust frameworks (Mayer 1995; TAM), ending in data-backed **AI-agent design cards**.

5 Limitations

- Trustpilot skews positive (opt-in bias) — we down-sample for classification.
- Equipment-rental platforms (Lensrentals, KitSplit) have few reviews -> weak per-platform claims there.
- One large r/ChatGPT thread pulls AI-side topics toward “AI tools”; few Reddit threads overall (38) cap LDA at 3–4 topics per theme.
- “Trust” is mostly *implicit* (host/renter, payment, review language), measured via topics and sentiment, not keyword counts. English-only.
- Reddit’s anti-bot wall blocks fresh anonymous re-scrapes (needs an OAuth app; see `legacy/`). Turo/Getaround per-platform review dates need a parse check in `02_merge_trustpilot.R` before any per-platform time-series.

Repository: <https://github.com/tyomachkaa/ai-acceptance-sharing-economy> · Online Content Analysis, WU Vienna, 2026.